

R Workshop – Day 4 – Probability, Sampling, Simulation and Inference

Applied R for Statistical Teaching, Research and Reproducible Analysis

Contents

1	R Workshop – Day 4 – Probability, Sampling, Simulation and Inference	2
1.1	Learning objectives	2
1.2	Capstone link	2
1.3	Segment plan (3 hours)	2
1.4	Part 1 – Concept bridge (Core workshop activity)	2
1.4.1	The <code>d/p/q/r</code> pattern	2
1.4.2	Why <code>set.seed()</code>	3
1.4.3	Sampling variation and the Central Limit Theorem (CLT)	3
1.5	Part 2 – Guided coding (Core workshop activity)	3
1.5.1	2.1 One simulation experiment	3
1.5.2	2.2 Demonstrating the CLT with repeated samples	3
1.5.3	2.3 One confidence interval	4
1.5.4	2.4 Selecting and running one test	4
1.5.5	2.5 Reporting statistical and practical significance	4
1.6	Part 3 – Participant practice (Core workshop activity)	5
1.7	Optional demonstration	5
1.8	Reference or take-home material	5
1.9	Daily deliverable	5
1.9.1	Evidence log entry	6
1.10	Facilitator notes	6

1 R Workshop – Day 4 – Probability, Sampling, Simulation and Inference

Session 4 of 7 | Duration: 3 hours Programme: Applied R for Statistical Teaching, Research and Reproducible Analysis

1.1 Learning objectives

By the end of this session you will be able to:

1. Use the **d/p/q/r** distribution-function pattern for at least the normal and binomial distributions.
2. Set a random seed and explain why it matters for reproducible simulation.
3. Run a simulation experiment and demonstrate the Central Limit Theorem with repeated sampling.
4. Estimate and interpret a confidence interval in plain language.
5. Choose an appropriate test based on study design and run it on the anchor dataset.
6. Report statistical and practical significance without overstating certainty.

1.2 Capstone link

Today produces one inferential answer to your capstone research question: typically, whether post-training scores improved relative to pre-training scores, and/or whether scores differ between two groups (e.g. completers versus non-completers).

1.3 Segment plan (3 hours)

Segment	Time	Purpose
Opening and recap	10 min	Recap Day 3 graphics, preview today's inferential question
Concept bridge	25 min	Probability functions, sampling variation, the CLT
Guided coding	60 min	Simulation, confidence interval, one hypothesis test
Participant practice	45 min	Run inference on the anchor dataset
Interpretation and reporting	25 min	Plain-language statistical reporting
Evidence log and capstone update	15 min	Save the inference notebook/script

1.4 Part 1 – Concept bridge (Core workshop activity)

1.4.1 The d/p/q/r pattern

Every distribution in R follows a naming convention, illustrated for the normal distribution:

Prefix	Meaning	Example
d	density (height of the curve at x)	<code>dnorm(70, mean = 65, sd = 10)</code>
p	cumulative probability, $P(X \leq x)$	<code>pnorm(70, mean = 65, sd = 10)</code>

Prefix	Meaning	Example
q	quantile, the x for a given cumulative probability	<code>qnorm(0.95, mean = 65, sd = 10)</code>
r	random draws from the distribution	<code>rnorm(10, mean = 65, sd = 10)</code>

The same four letters apply to `binom` (binomial), `unif` (uniform), `t` (t-distribution), and others.

1.4.2 Why `set.seed()`

```
set.seed(2024)
rnorm(3)
```

Running the line above twice with the same seed gives the same three numbers. Without `set.seed()`, every run of a simulation differs, which makes results impossible to check or reproduce. Set the seed once, near the top of the script, before the first random draw.

1.4.3 Sampling variation and the Central Limit Theorem (CLT)

A single sample mean is one estimate of the population mean; a different sample would give a different estimate. The CLT says that, as sample size grows, the distribution of the *sample mean* across repeated samples becomes approximately normal, regardless of the shape of the original population – which is why so much of classical inference relies on the normal distribution.

1.5 Part 2 – Guided coding (Core workshop activity)

```
library(tidyverse)

lecturers <- read_csv("Data/processed/anchor_dataset_clean.csv", show_col_types =
  ↪ FALSE)
set.seed(2024)
```

1.5.1 2.1 One simulation experiment

```
# Simulate 1,000 lecturers' post-training scores under an assumed
# population mean of 65 and sd of 12, purely to see typical sample noise.
simulated_scores <- rnorm(1000, mean = 65, sd = 12)
mean(simulated_scores)
sd(simulated_scores)
hist(simulated_scores, main = "Simulated post-training scores", xlab = "Score")
```

1.5.2 2.2 Demonstrating the CLT with repeated samples

```
sample_means <- replicate(2000, {
  one_sample <- sample(lecturers$post_score[!is.na(lecturers$post_score)], size =
  ↪ 30)
  mean(one_sample)
```

```

}))

hist(sample_means, main = "Distribution of sample means (n = 30, 2000 repeats)",
      xlab = "Sample mean post-training score")
sd(sample_means) # the standard error, empirically

```

Compare `sd(sample_means)` to `sd(lecturers$post_score, na.rm = TRUE) / sqrt(30)` – the two should be close, which is the CLT’s standard-error formula made concrete.

1.5.3 2.3 One confidence interval

```

post_scores <- lecturers$post_score[!is.na(lecturers$post_score)]
n <- length(post_scores)
sample_mean <- mean(post_scores)
sample_se <- sd(post_scores) / sqrt(n)

ci_lower <- sample_mean - qt(0.975, df = n - 1) * sample_se
ci_upper <- sample_mean + qt(0.975, df = n - 1) * sample_se
c(ci_lower, ci_upper)

# Or, directly from a t-test:
t.test(post_scores)$conf.int

```

Plain-language template: “We are 95% confident that the true mean post-training score for the population this sample represents lies between `ci_lower` and `ci_upper`.”

1.5.4 2.4 Selecting and running one test

Test choice follows study design, not habit:

Design	Typical test
One numeric variable vs. a fixed value	One-sample t-test
Same individuals measured twice (pre/post)	Paired t-test
Two independent groups, one numeric outcome	Independent (two-sample) t-test
Two categorical variables	Chi-square test of independence

For the capstone question – did scores improve from pre- to post-training – the design is paired:

```

paired_data <- lecturers %>% filter(!is.na(pre_score), !is.na(post_score))

t.test(paired_data$post_score, paired_data$pre_score, paired = TRUE)

```

1.5.5 2.5 Reporting statistical and practical significance

```

paired_result <- t.test(paired_data$post_score, paired_data$pre_score, paired =
  ↪ TRUE)
mean_gain <- mean(paired_data$post_score - paired_data$pre_score)

```

Reporting template: “Post-training scores were on average `mean_gain` points higher than pre-training scores (paired t-test, $t = \dots$, $df = \dots$, $p = \dots$). This difference is [statistically significant /

not statistically significant] at the 5% level, and represents a [meaningful / modest] practical gain on a 0-100 scale.”

Statistical significance (a small p-value) and practical significance (a gain large enough to matter to a lecturer or programme manager) are different claims – report both.

1.6 Part 3 – Participant practice (Core workshop activity)

1. Compute a 95% confidence interval for mean attendance.
2. Run the paired t-test for pre- versus post-training scores and write the plain-language sentence above with your actual numbers filled in.
3. Choose one additional comparison relevant to your research question (e.g. post-training score for completers versus non-completers, an independent two-sample t-test) and run it.
4. For your chosen test, check at least one assumption informally (e.g. a histogram or `qqnorm()` of the relevant variable) and note any concern in a comment.
5. Save your code and your written interpretation as `Outputs/day4_inference_notebook.R` (or an `.Rmd/.qmd` if you prefer).

A facilitator-reviewed solution is in `Solution-Scripts/day4_inference_solution.R`.

1.7 Optional demonstration

- **Coverage simulation** – repeat the confidence-interval calculation across many simulated samples and check what proportion actually contain the true mean (should be close to 95%).
- **Chi-square and Fisher’s exact test** – e.g. is `completed` associated with `training_track`?
- **Non-parametric alternatives** – Wilcoxon signed-rank test as a paired-t-test alternative when normality is doubtful.
- **Effect-size calculations** – Cohen’s d via the `effectsize` package.
- **Additional CLT demonstrations** – repeat Part 2.2 with a clearly skewed variable (e.g. `years_teaching`) to show the CLT still holds for the *sample mean* even when the raw variable is not normal.

```
# Optional: chi-square test of independence
chisq.test(table(lecturers$completed, lecturers$training_track))
```

1.8 Reference or take-home material

- Assumption checklists for the t-test family (independence, approximate normality of the relevant quantity, for two-sample tests: variance assumptions and Welch’s correction).
- A short template for reporting a p-value and confidence interval consistently across the rest of the workshop.

1.9 Daily deliverable

A short inference notebook or script (`Outputs/day4_inference_notebook.R` or equivalent) that answers one inferential question linked to the capstone research question, with a plain-language interpretation.

1.9.1 Evidence log entry

Complete the Day 4 row in `Workshop-Evidence-Log.md`: state your inferential result and whether it supports, contradicts, or is inconclusive about your initial impression from the Day 3 graphics.

1.10 Facilitator notes

- The paired t-test on `pre_score/post_score` should show a positive, statistically significant gain by construction of the synthetic dataset – use this as your answer key for spot-checking participants' work.
- Watch for participants computing a confidence interval or t-test on a column that still contains `NA` without filtering first; `t.test()` will error or silently drop rows depending on the call, so this is worth flagging explicitly.
- If a participant chooses chi-square as their “one additional comparison,” that is acceptable even though it is listed as optional demonstration – the boundary exists for time management, not as a hard rule.